

Q1.

A uniform solid is formed by rotating the region enclosed by the positive x -axis, the line x

$= 2$ and the curve $y = \frac{1}{2}x^2$ through 360° around the x -axis.

(a) Find the centre of mass of this solid.

(5)

(b) The solid is placed with its plane face on a rough inclined plane and does not slide.

The angle between the inclined plane and the horizontal is gradually increased.

When the angle between the inclined plane and the horizontal is α , the solid is on the point of toppling.

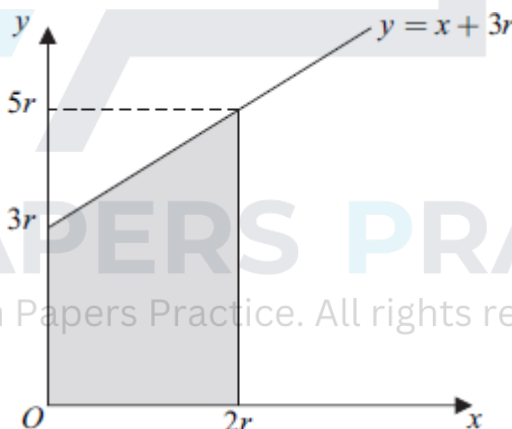
Find α , giving your answer to the nearest 0.1°

(2)

(Total 7 marks)

Q2.

The region bounded by the line $y = x + 3r$, the y -axis, the x -axis and the line $x = 2r$ is shown in the diagram.

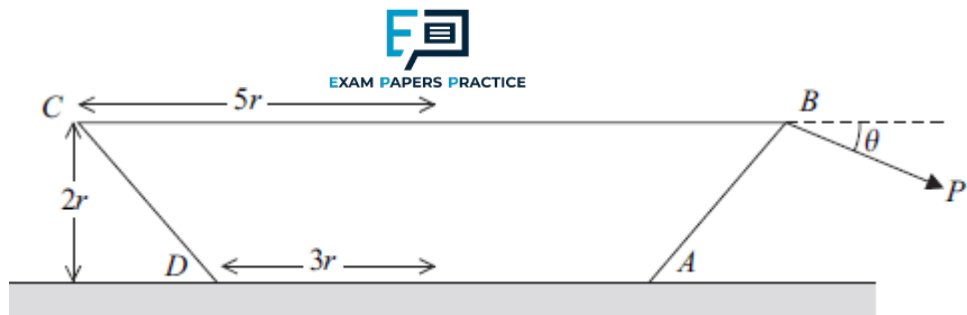


This region is rotated about the x -axis to form a frustum, of volume $\frac{98\pi r^3}{3}$, of a uniform solid cone.

(a) Using integration, find the distance of the centre of mass of the frustum of the cone from O .

(6)

(b) The frustum of a uniform solid cone with radii $3r$ and $5r$ and height $2r$ has weight W . The frustum stands on a horizontal surface. The diagram shows a cross-section, $ABCD$, which includes the axis of symmetry, of this frustum. A force P is applied to the point B and acts in the same plane as this cross-section at an angle of θ below the horizontal.



The frustum does not slide and is about to topple about A.

- (i) Show that

$$P = \frac{3W}{2(\cos \theta + \sin \theta)}$$

(4)

- (ii) Find, in terms of W , the minimum possible value of P .

(3)

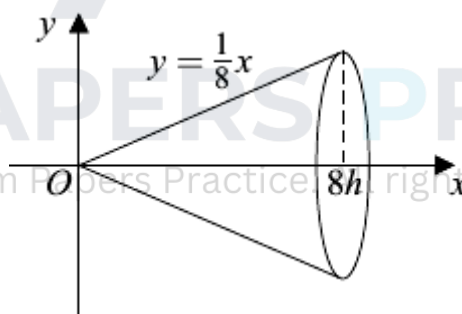
- (iii) State the value of θ for which P is a minimum.

(1)

(Total 14 marks)

Q3.

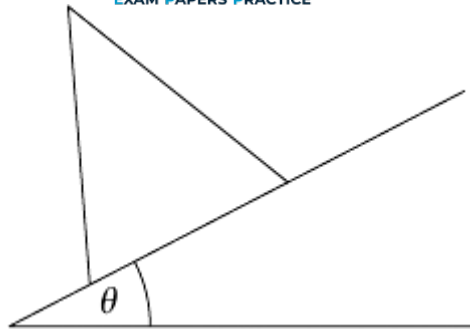
A uniform solid cone is formed by rotating the finite region bounded by the lines $y = \frac{1}{8}x$, $y = 0$ and $x = 8h$ through 2π radians about the x -axis. The cone is shown in the diagram.



- (a) Use integration to show that the distance of the centre of mass of the cone from the origin is $6h$.

(5)

- (b) The cone rests in equilibrium with its plane face on a rough plane inclined at an angle θ to the horizontal, as shown in the diagram.



- (i) On the diagram, show all the forces acting on the cone.

(2)

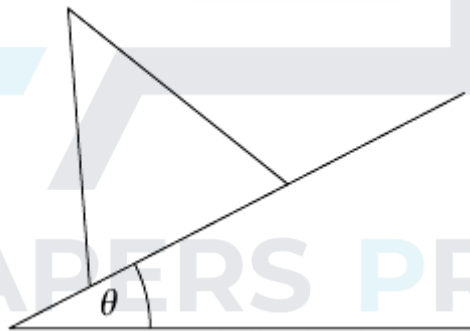
- (ii) If the plane is sufficiently rough to prevent slipping, find the maximum value of θ for the cone to remain in equilibrium without toppling.

(3)

- (c) The cone is now placed with its plane face on another rough inclined plane. The coefficient

of friction between the cone and this plane is $\frac{6}{13}$. The angle α between the inclined plane and the horizontal is gradually increased from $\alpha = 0^\circ$.

Determine whether the cone will topple first or slide first.



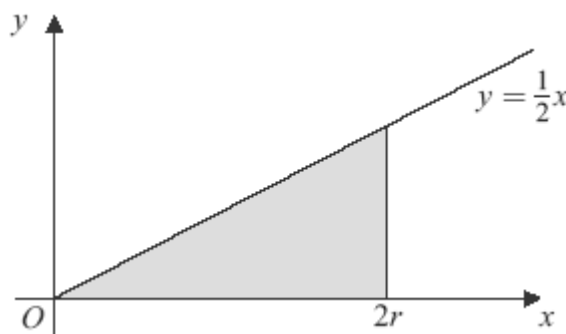
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(4)

(Total 14 marks)

Q4.

The region bounded by the line $y = \frac{1}{2}x$, the x -axis and the line $x = 2r$ is shown in the diagram.

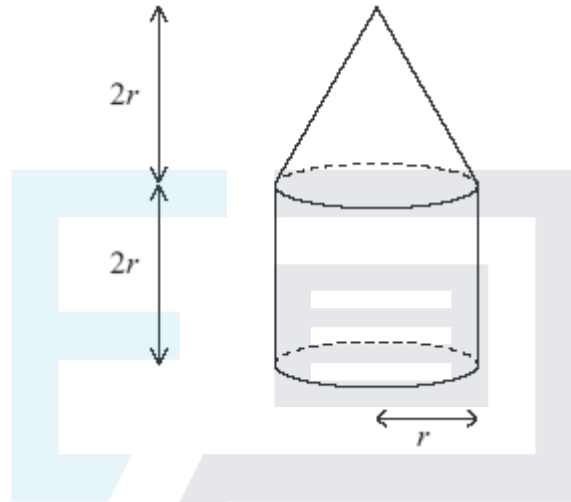


This region is rotated about the x -axis to form a uniform solid cone of height $2r$ and radius r .

- (a) Show, using integration, that the centre of mass of the cone is at a distance of $\frac{3r}{2}$ from the origin.

(5)

- (b) A rocket consists of two parts. The lower part of the rocket may be modelled as a uniform solid cylinder with radius r , height $2r$ and density ρ . The upper part of the rocket may be modelled as a uniform solid cone of radius r , height $2r$ and density $k\rho$, as shown in the diagram.

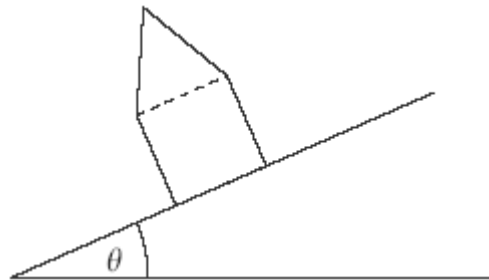


- (i) Show that the centre of mass of the rocket is at a distance of $\left(\frac{6+5k}{6+2k}\right)r$ from the base of the rocket.

(5)

- (ii) The rocket is now placed on a rough plane, which is inclined at an angle of θ to the horizontal, where $\tan \theta = \frac{2}{3}$.

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Given that the rocket does **not** slide and is just on the point of toppling, find the value of k .

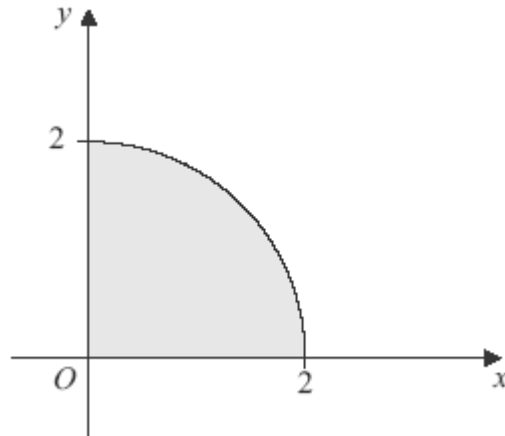
(5)

(Total 15 marks)

Q5.

The region bounded by the positive x -axis, the positive y -axis and the curve with equation

$y = \sqrt{4 - x^2}$ is shown in the diagram.



The region is rotated through 2π radians around the x -axis to form a uniform solid.

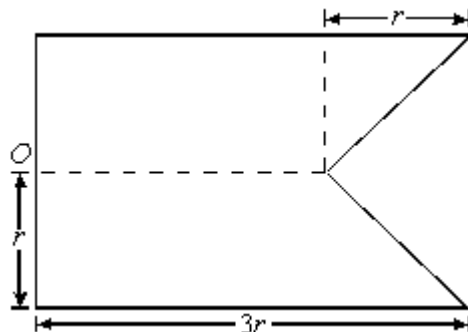
- (a) Use integration to show that the volume of the solid is $\frac{16\pi}{3}$. (3)
- (b) Use integration to find the distance of the centre of mass of the solid from the y -axis. (4)
- (c) The solid is suspended from a point on the edge of its circular face and hangs in equilibrium. (3)
- Find the angle between the circular face and the vertical.

(Total 10 marks)

Q6.

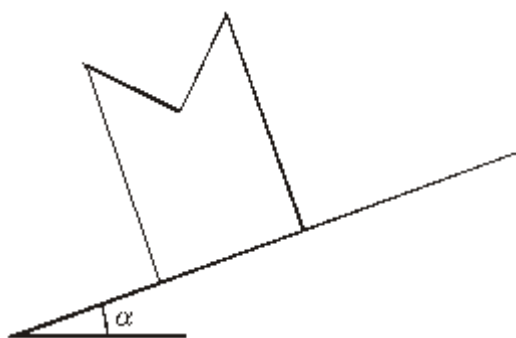
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A solid, uniform cylinder, of radius r and length $3r$, has a cone of radius r and height r drilled out of one end. The diagram shows a cross section along the axis of symmetry of the body. Point O is the centre of the plane circular surface of the body.



- (a) Show that the centre of mass of the body is $\frac{43r}{32}$ from point O . (6)

- (b) The body is placed with its plane surface on an adjustable rough slope as shown. The angle α which the plane makes with the horizontal is gradually increased from zero.



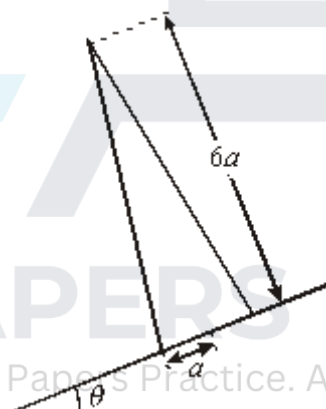
Assuming that the body does **not** slide, find, to the nearest degree, the value of angle α when the body just begins to topple.

(3)

(Total 9 marks)

Q7.

A uniform solid cone of weight W rests in equilibrium with its plane face on a rough plane inclined at an angle θ to the horizontal. The cone is of height $6a$ and the base radius a .



- (a) Draw a diagram showing the forces acting on the cone.

(2)

- (b) The angle θ is gradually increased from $\theta = 0^\circ$.

Assuming that the plane is sufficiently rough to prevent slipping, show that the cone is about to topple when $\tan \theta = \frac{2}{3}$.

(3)

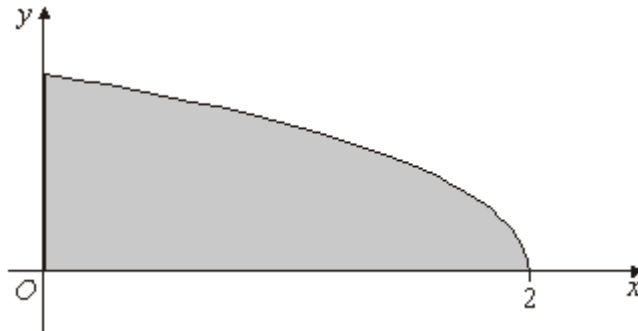
- (c) The cone is now placed on another rough inclined plane where the coefficient of friction between the cone and the plane is 0.6. The angle θ between the plane and the horizontal is gradually increased from $\theta = 0^\circ$. Determine whether the cone will topple or slide first.

(6)

(Total 11 marks)

Q8.

The region bounded by the x -axis, the y -axis and the curve with equation $y = \sqrt{2-x}$ is shown in the diagram below.

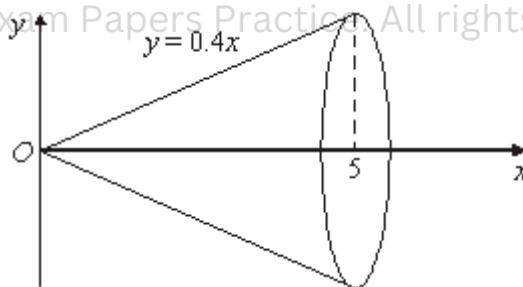


The region is rotated through 360° around the x -axis to form a uniform solid.

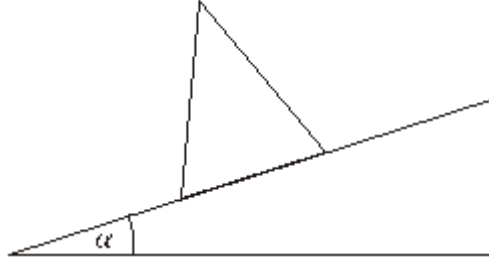
- (a) Show that the volume of the solid is 2π . (3)
 - (b) Find the distance of the centre of mass of the solid from the y -axis. (4)
 - (c) The solid is suspended from a point on the edge of its circular face and hangs in equilibrium. Find the angle between the circular face and the vertical. (4)
- (Total 11 marks)**

Q9.

A uniform **solid** cone is formed by rotating the finite region bounded by the lines with equations $y = 0.4x$, $y = 0$ and $x = 5$ through 360° about the x -axis. The cone is shown in the diagram below.



- (a) Use integration to show that the distance of the centre of mass of the cone from the origin is $\frac{15}{4}$. (5)
- (b) The cone is placed on a rough slope, which is at an angle α to the horizontal, as shown in the diagram.



The angle α is gradually increased.

Assuming that the cone does not slide, find the value of α when the cone is on the point of toppling.

(5)

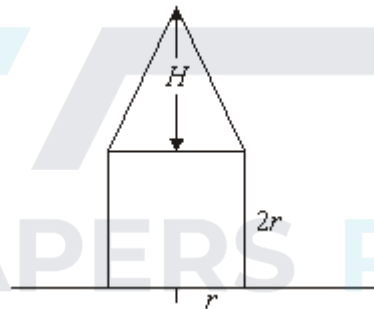
(Total 10 marks)

Q10.

The diagram shows a cross-section of a child's toy, which consists of a uniform solid cylinder and a uniform solid right circular cone. One plane face of the cylinder is fixed to the plane face of the cone.

The cylinder has density ρ , radius r and height $2r$.

The cone has density 3ρ , height H and radius r .

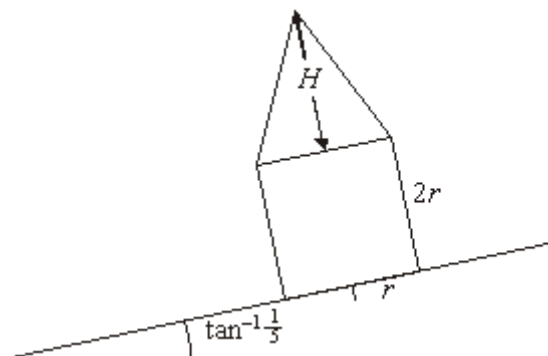


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- (a) Show that the centre of mass of the toy is $\frac{8r^2 + 8rH + H^2}{4(2r + H)}$ from its base.

(6)

- (b) The toy is now placed on a rough plane inclined at an angle of $\tan^{-1} \frac{1}{5}$ to the horizontal.



Show that the maximum value of H if the toy is not to topple is $(6 + 2\sqrt{17})r$.

(5)

(Total 11 marks)



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